ALONG AN EFFECTIVE CARTIER DIVISOR ON A REGULAR SURFACE THE DETERMINANT OF AN EXACT SEQUENCE OF RANK R SHEAVES CHANGES BY A BOUNDED TWIST

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-wedge-product-ses is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 3276-3295 and proof span 3297-3331 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains a displayed choice between two cited structure theorems for modules over a discrete valuation ring (an 'or' between two labels); the author's wording and both labels are preserved, and both are handled as external navigation pointers.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Reflexive sheaves over a regular scheme of dimension two on a nonempty open where a coherent sheaf is locally free of rank r have invertible determinant, cited from the same chapter. Cited at proof line 3298. Makes the two determinants invertible, so that the canonical map between them can be compared as a section of an invertible sheaf.
- The structure theorem for finite modules over a discrete valuation ring (or over a PID), cited from the more-algebra chapter, and the fact that dividing a finite module by its torsion does not change its double dual. Cited at proof lines 3312-3315 and used at 3316-3322. Reduces the algebraic problem to free modules of rank r, where the determinant is available.
- The vanishing order of the determinant of a map between free modules equals the length of the cokernel, cited from the algebra chapter. Cited at proof lines 3329-3331. Gives the exact vanishing order s of the determinant in the reduced case, which is at most the minimum of s and r.

Source reference. Original lemma algebra-lemma-order-vanishing-determinant: sources/stacks-divisors/pinned-original-source.tex, source lines 3330–3330. Same-chapter reference

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used in the retained author text at source line(s) 3330. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0B3N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13971–13971. Same-chapter reference used in the retained author text at source line(s) 3298. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma more-algebra-lemma-generalized-valuation-ring-modules: sources/stacks-divisors/pinned-original-source.tex, source lines 3319–3319. Same-chapter reference used in the retained author text at source line(s) 3319. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma more-algebra-lemma-modules-PID: sources/stacks-divisors/pinned-original-source.tex, source lines 3320–3320. Same-chapter reference used in the retained author text at source line(s) 3320. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be an integral regular scheme of dimension 2. Let $i : D \to X$ be the immersion of an effective Cartier divisor. Let $\mathcal{F} \to \mathcal{F}' \to i_*\mathcal{G} \to 0$ be an exact sequence of coherent $\mathcal{O}_X$-modules. Assume*

(1) *$\mathcal{F}, \mathcal{F}'$ are locally free of rank r on a nonempty open of X,*
(2) *D is an integral scheme,*
(3) *$\mathcal{G}$ is a finite locally free $\mathcal{O}_D$-module of rank s.*

Then $\mathcal{L} = (\wedge^r \mathcal{F})^{**}$ and $\mathcal{L}' = (\wedge^r \mathcal{F}')^{**}$ are invertible $\mathcal{O}_X$-modules and $\mathcal{L}' \cong \mathcal{L}(kD)$ for some $k \in \{0, \ldots, \min(s, r)\}$.

Proof. The first statement follows from Lemma 0B3N as assumption (1) implies that $\mathcal{L}$ and $\mathcal{L}'$ have rank 1. Taking $\wedge^r$ and double duals are functors, hence we obtain a canonical map $\sigma : \mathcal{L} \to \mathcal{L}'$ which is an isomorphism over the nonempty open of (1), hence nonzero. To finish the proof, it suffices to see that σ viewed as a global section of $\mathcal{L}' \otimes \mathcal{L}^{\otimes -1}$ does not vanish at any codimension point of X, except at the generic point of D and there with vanishing order at most $\min(s, r)$.

Translated into algebra, we arrive at the following problem: Let $(A, \mathfrak{m}, \kappa)$ be a discrete valuation ring with fraction field K. Let $M \to M' \to N \to 0$ be an exact sequence of finite A-modules with $\dim_K(M \otimes K) = \dim_K(M' \otimes K) = r$ and with $N \cong \kappa^{\oplus s}$. Show that the induced map $L = \wedge^r(M)^{**} \to L' = \wedge^r(M')^{**}$ vanishes to order at most $\min(s, r)$. We will use the structure theorem for modules over A, see More on Algebra, Lemma more-algebra-lemma-generalized-valuation-ring-modules or more-algebra-lemma-modules-PID. Dividing out a finite A-module by a torsion submodule does not change the double dual. Thus we may replace M by M/M_{tors} and M' by $M'/\operatorname{Im}(M_{tors} \to M')$ and assume that M is torsion free. Then $M \to M'$ is injective and $M'_{tors} \to N$ is injective. Hence we may replace M' by M'/M'_{tors} and N by N/M'_{tors}. Thus we reduce to the case where M and M' are free of rank r and $N \cong \kappa^{\oplus s}$. In this case σ is the determinant of $M \to M'$ and vanishes to order s for example by Algebra, Lemma algebra-lemma-order-vanishing-determinant. □